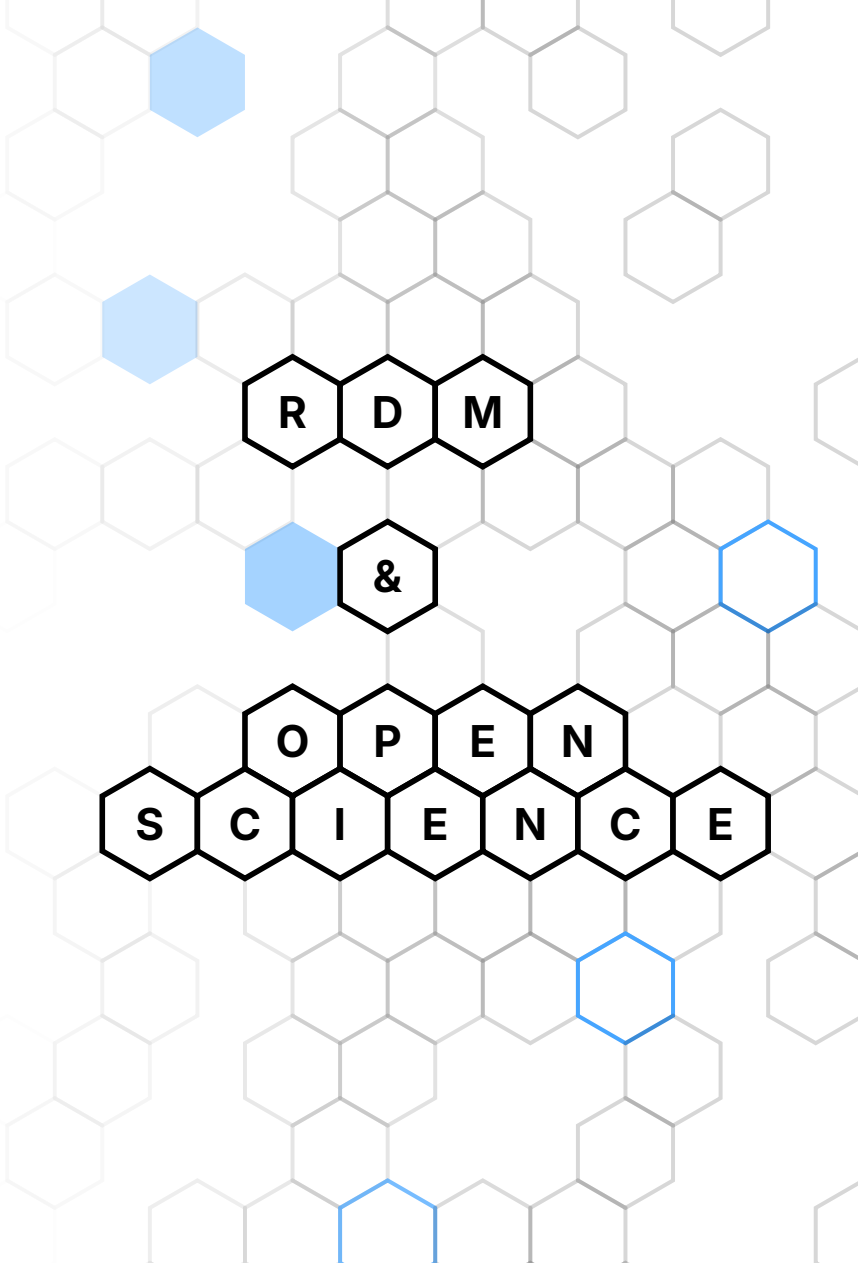
Aalto-yliopisto
Aalto-universitetet
Aalto University

# Image generation with AI

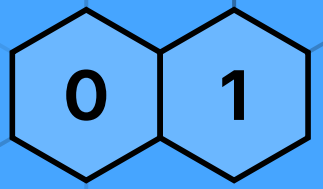
AI in research work

**Pedro Santos Silva, PhD · Research Software Engineer**

Aalto University · 22.10.2026



# Why use AI to generate images?




A!

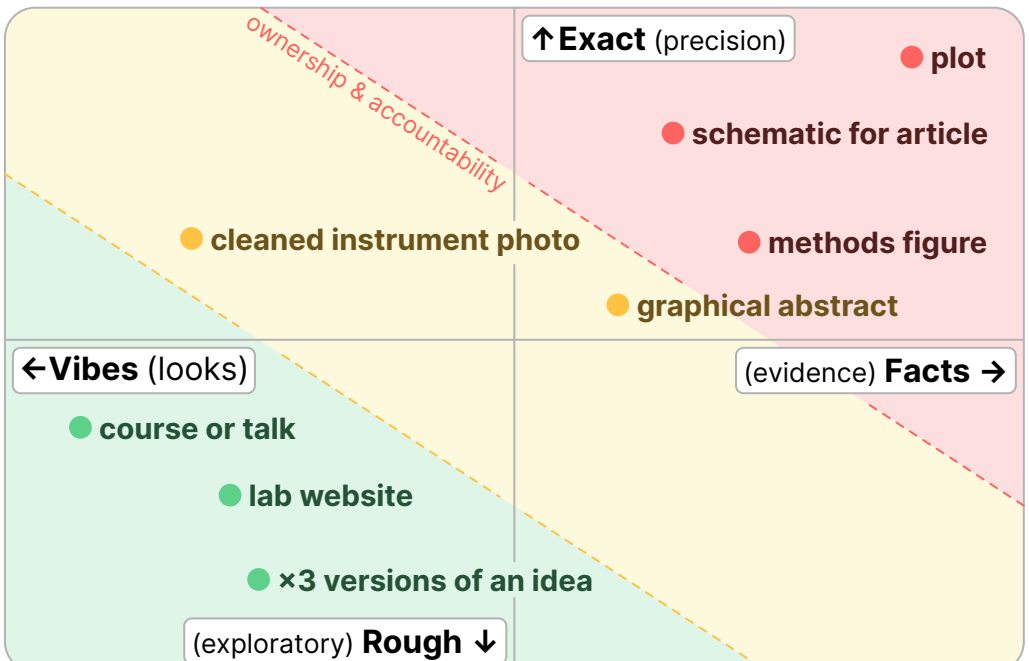
# When do you need to use images in your research work?

I need an image for...

- the lab website
- visualize an idea
- my showcase my data
- a course or a talk
- drawing my own setup
- a research paper

Show of hands — who recognised one of these?

# Where each one sits




**Effort level**

**1. USE IT**
**Prompt & publish**
Safe to use directly & exploratory. Use AI output as-is.

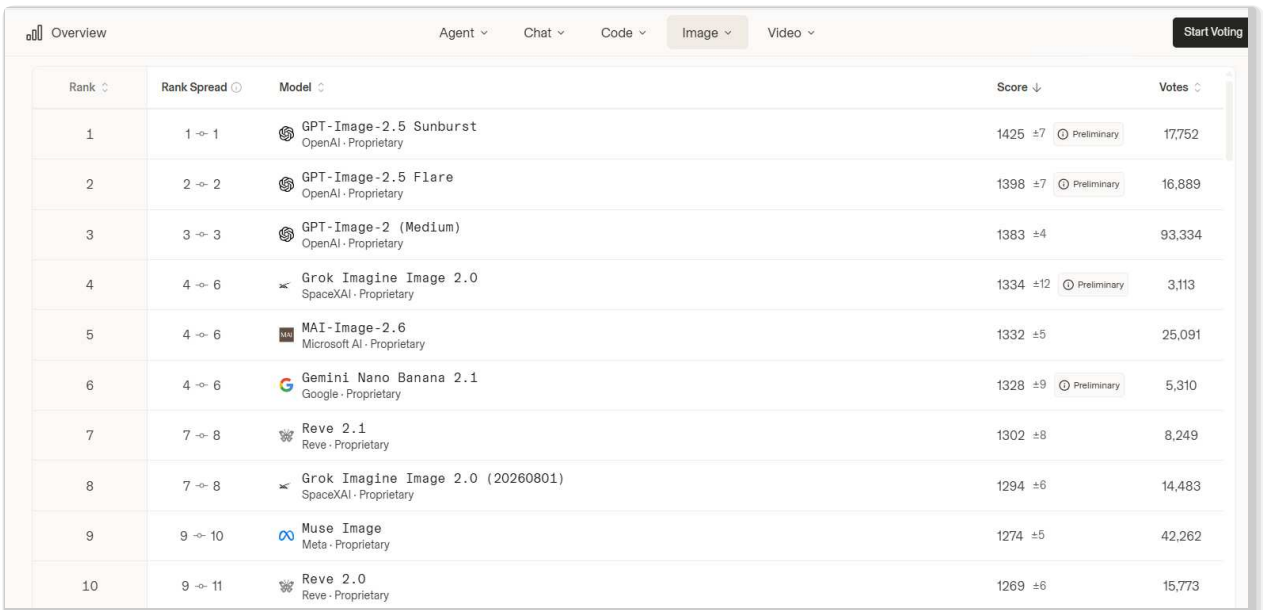
**2. FINISH IT**
**AI base + human edit**
Use AI for layout or cleanup, but verify & refine details yourself.

**3. DO IT YOURSELF**
**100% human ownership**
AI can help with layout or cleanup, but the final image needs to be human generated. Full accountability.

**Good practices:** Test ideas with AI. Prove facts with code and data. If you want to be the owner of the final image, you need to put effort into creating it.

Images © 2026 Pedro Silva. Licensed under CC BY 4.0.

# Which tool should you use? (as of November 2026)



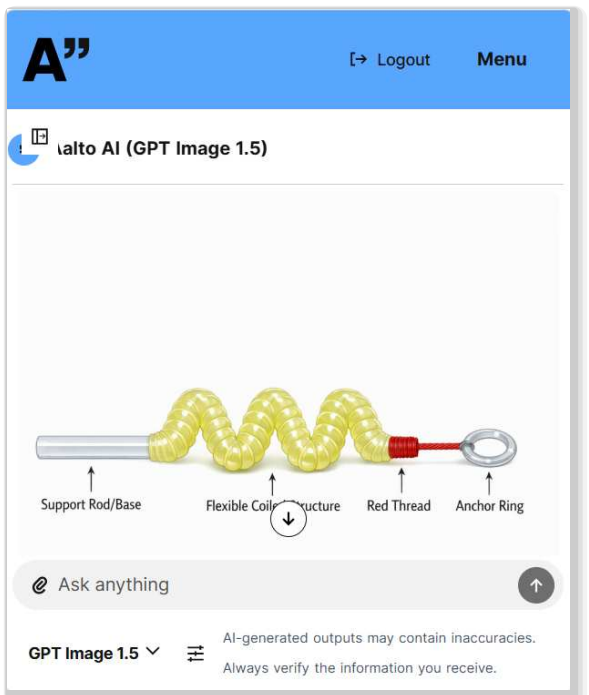

| Rank | Rank Spread | Model | Score | Votes |
|---|---|---|---|---|
| 1 | 1 → 1 | GPT-Image-2.5 Sunburst<br>OpenAI · Proprietary | 1425 ±7 Preliminary | 17,752 |
| 2 | 2 → 2 | GPT-Image-2.5 Flare<br>OpenAI · Proprietary | 1398 ±7 Preliminary | 16,889 |
| 3 | 3 → 3 | GPT-Image-2 (Medium)<br>OpenAI · Proprietary | 1383 ±4 | 93,334 |
| 4 | 4 → 6 | Grok Imagine Image 2.0<br>SpaceXAI · Proprietary | 1334 ±12 Preliminary | 3,113 |
| 5 | 4 → 6 | MAI-Image-2.6<br>Microsoft AI · Proprietary | 1332 ±5 | 25,091 |
| 6 | 4 → 6 | Gemini Nano Banana 2.1<br>Google · Proprietary | 1328 ±9 Preliminary | 5,310 |
| 7 | 7 → 8 | Reve 2.1<br>Reve · Proprietary | 1302 ±8 | 8,249 |
| 8 | 7 → 8 | Grok Imagine Image 2.0 (20260801)<br>SpaceXAI · Proprietary | 1294 ±6 | 14,483 |
| 9 | 9 → 10 | Muse Image<br>Meta · Proprietary | 1274 ±5 | 42,262 |
| 10 | 9 → 11 | Reve 2.0<br>Reve · Proprietary | 1269 ±6 | 15,773 |

New models are added every day (hard to keep up-to-date)

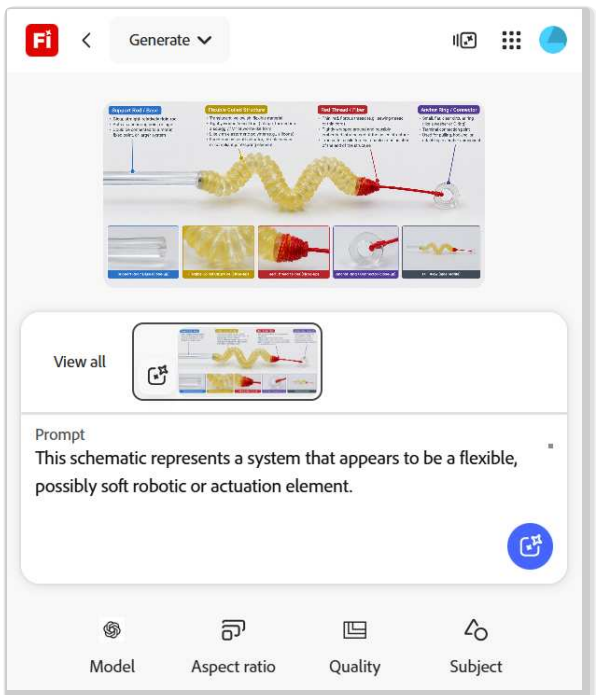
# How to use the AI tools available at Aalto



# Easy access tools available at Aalto




ai.aalto.fi




Adobe's Firefly

# What is a good prompt?

**Quick start:**

a) Check the tool's official documentation for the latest prompting recommendations.
b) Use meta-prompts when possible (e.g., "Generate a prompt for model gpt-image-1.5 that will produce an image of…").

Example of prompt basics for Midjourney:

- **Subject**: Who or what? (person, animal, character, location, object)
- **Medium**: In what form? (photo, painting, illustration, sculpture, doodle, tapestry)
- **Environment**: Where? (indoors, outdoors, on the moon, underwater, in the city)
- **Lighting**: What kind? (soft, ambient, overcast, neon, studio lights)
- **Color**: In what shades? (vibrant, muted, bright, monochromatic, colorful, black and white, pastel)
- **Mood**: Feelings to evoke? (playful, calm, gloomy, energetic)
- **Composition**: How is it framed? (portrait, headshot, closeup, birds-eye view)

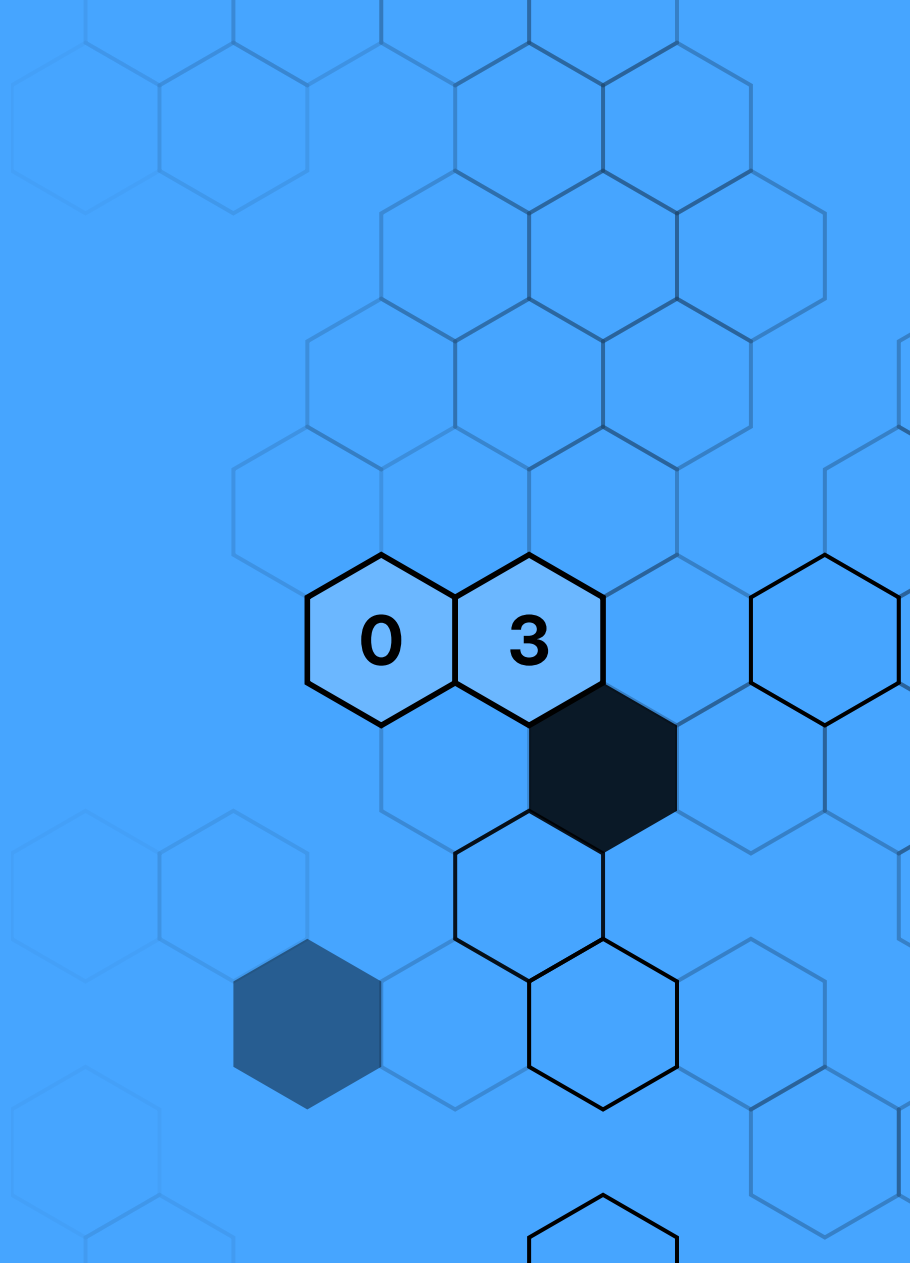
The model fills in the rest, so leave nothing important blank.

# Useful keywords for research image generation

| Prompt keywords to add | What it suggests the model to do | Best for / Watch out |
|---|---|---|
| scientific diagram · schematic illustration | Separates parts, flattens perspective, adds callout lines | **Process flows & setups** strip AI text & re-label yourself |
| flat vector illustration · iconographic | Kills photorealistic lighting, shadows, and lens blur | **Slide icons & graphical abstracts** cleanest background removal |
| minimalist line art · white background | High-contrast ink strokes only — a drawing, not a photo | **Whiteboard / xkcd concepts** easy to trace or threshold |
| isometric cutaway · orthographic 3D | Parallel projection where depth and hidden layers stay readable | **Hardware rigs & layer stacks** check port & screw counts |
| patent drawing · technical cross-section | Monochrome stippling and numbered leader lines on plain paper | **Mechanical assemblies** great when photos look too messy |
| Okabe-Ito palette · high contrast | Pushes distinct hues instead of muddy cinematic gradients | **Review-safe figures** still verify contrast in grayscale |

**Tip:** Avoid saying **make it look professional**, name the **medium** and the **projection** instead.

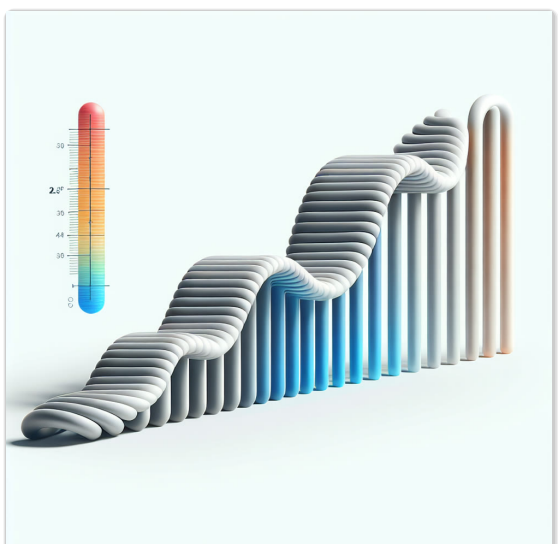
# Where to start?



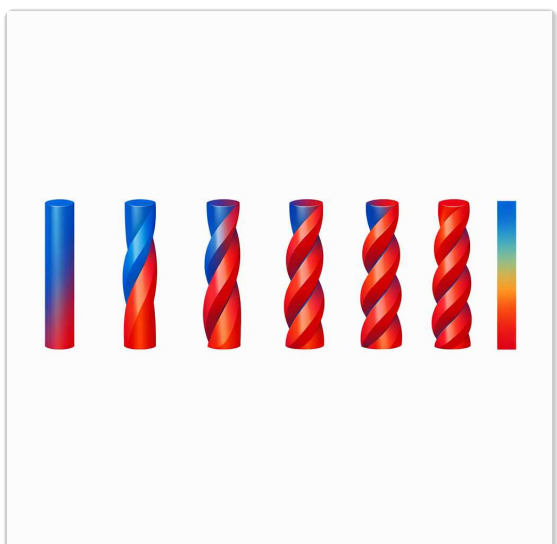
# From ideas to images

**PROMPT**

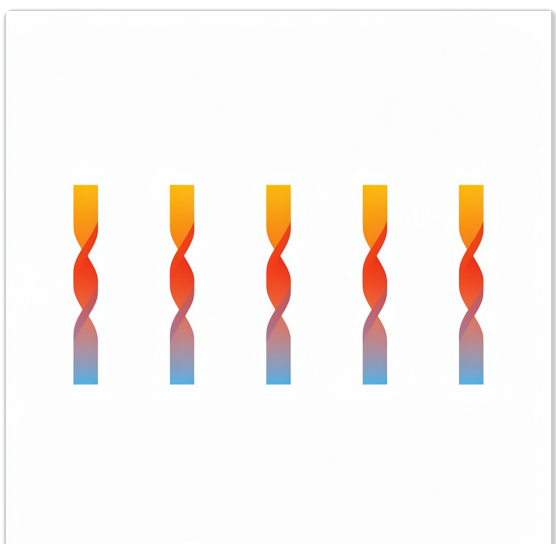
Scientific concept graphic on a pure white background: a left-to-right sequence of five identical rods, each gently twisting more than the previous, colored by a smooth low-to-high gradient along their length. Add a small vertical color bar at the right with no tick numbers or words. Flat vector look, crisp edges, no shadows, no textures, no photorealism.



Dall-E·3



GPT Image 1.5



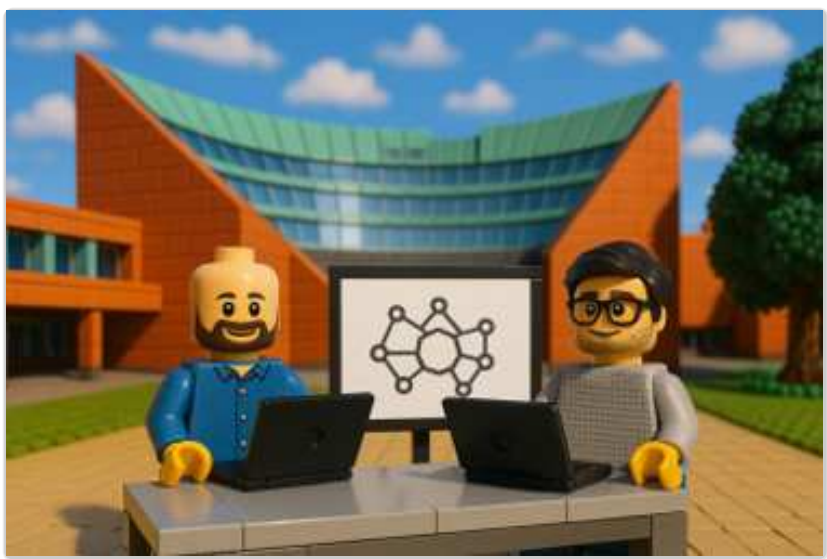
**Flux 2 Pro**

# Combining reference images

**PROMPT**

researcher_1.jpg researcher_2.jpg undergraduate_center.jpg

Draw two researchers (pictures attached) from Aalto University (the image attached contains a landmark building), giving a webinar about Prompt Engineering. The artistic style should resemble a Lego movie.



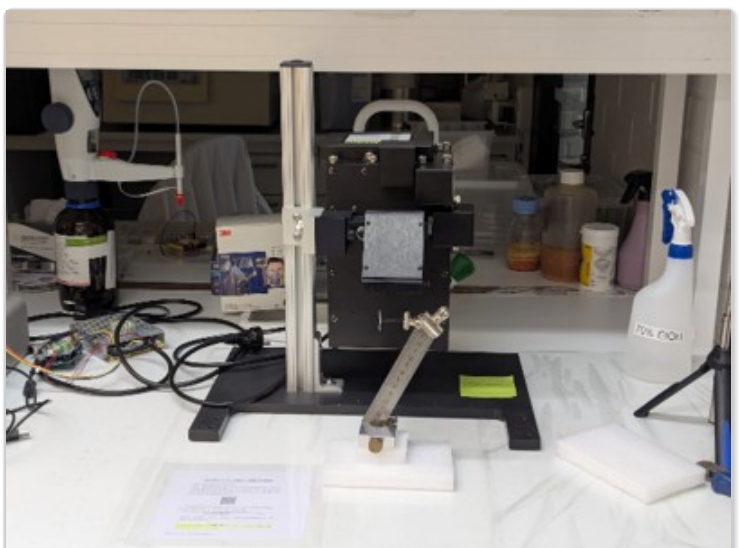
**GPT Image 1**

# Editing pictures

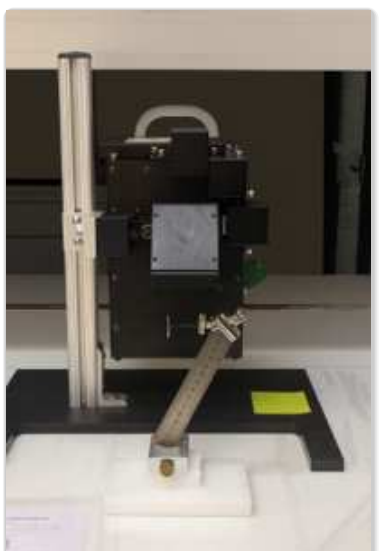
**PROMPT**

messy_solar_simulator.jpg

Remove all cables, bottles and chemicals from the image.



`messy_solar_simulator.jpg`



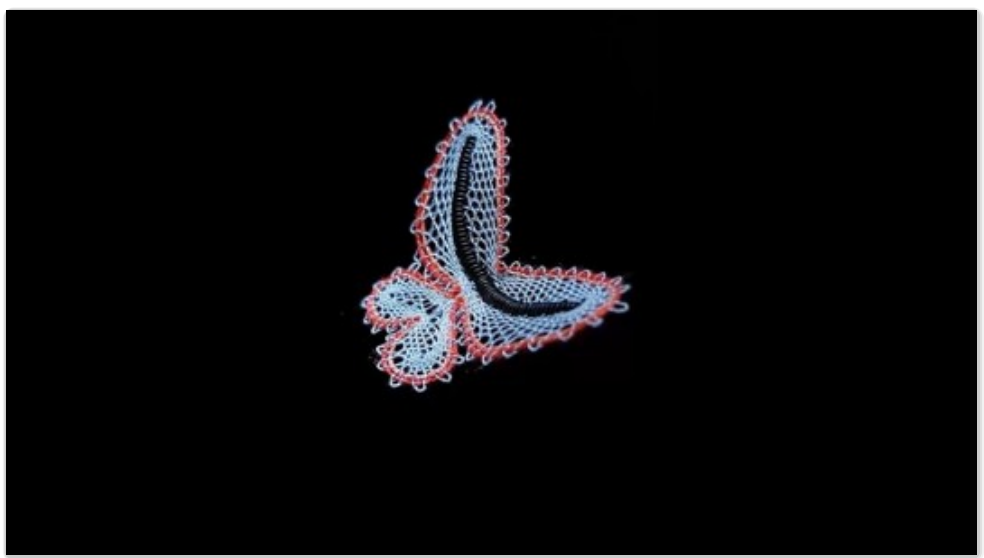
**GPT Image 1**

# Putting your own result in a context

**PROMPT**

butterfly-source.jpg

Generate a beautiful realistic nature, featuring the butterfly actuator living among real ones. It should look natural, calm and symbiotic.



butterfly-source.jpg



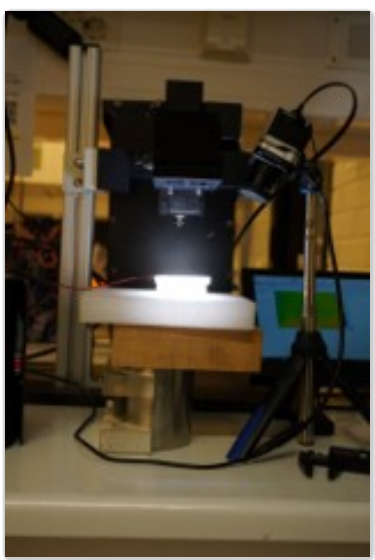
**GPT Image 1**

# From a photo of your setup to a schematic

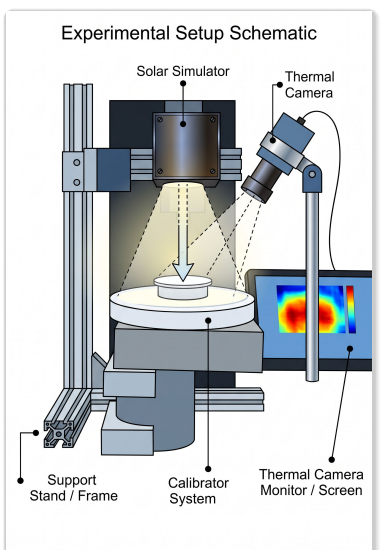
**PROMPT**

`messy_solar_simulator.jpg`

I've attached a picture of my experimental setup. I want you to make a representation in the style of a **scientific diagram**. My picture is a photo so it has **many elements that are not needed**. In the center top it has the solar simulator, the calibrator system is in the center and being illuminated, and in the right top there's the screen showing the image of the thermal camera and in the right there's the screen showing the image of the thermal camera.



`messy_solar_simulator.jpg`



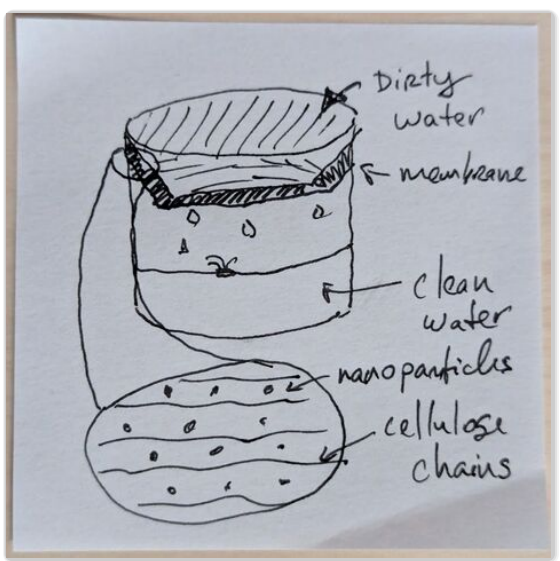

**Nano Banana 2.5**

# From a sketch to a full picture

**PROMPT**

`filter_sketch.jpg`

I want to make a diagram for my research article. It needs to have a good finishing without becoming sci-fi. **Generate a prompt** that I can use with this image so it takes full advantage of the latest image generation models of ChatGPT. The fidelity does not need to be 100% because I draw really bad, I tried to depict from top to bottom in the image dirty water on top of a glass, goes through a filter and in the bottom there's clean water. The left side starts in the filter and it's supposed to be a super ampliation to see the internal structure at micro/nano level of cellulose chains and nanoparticles

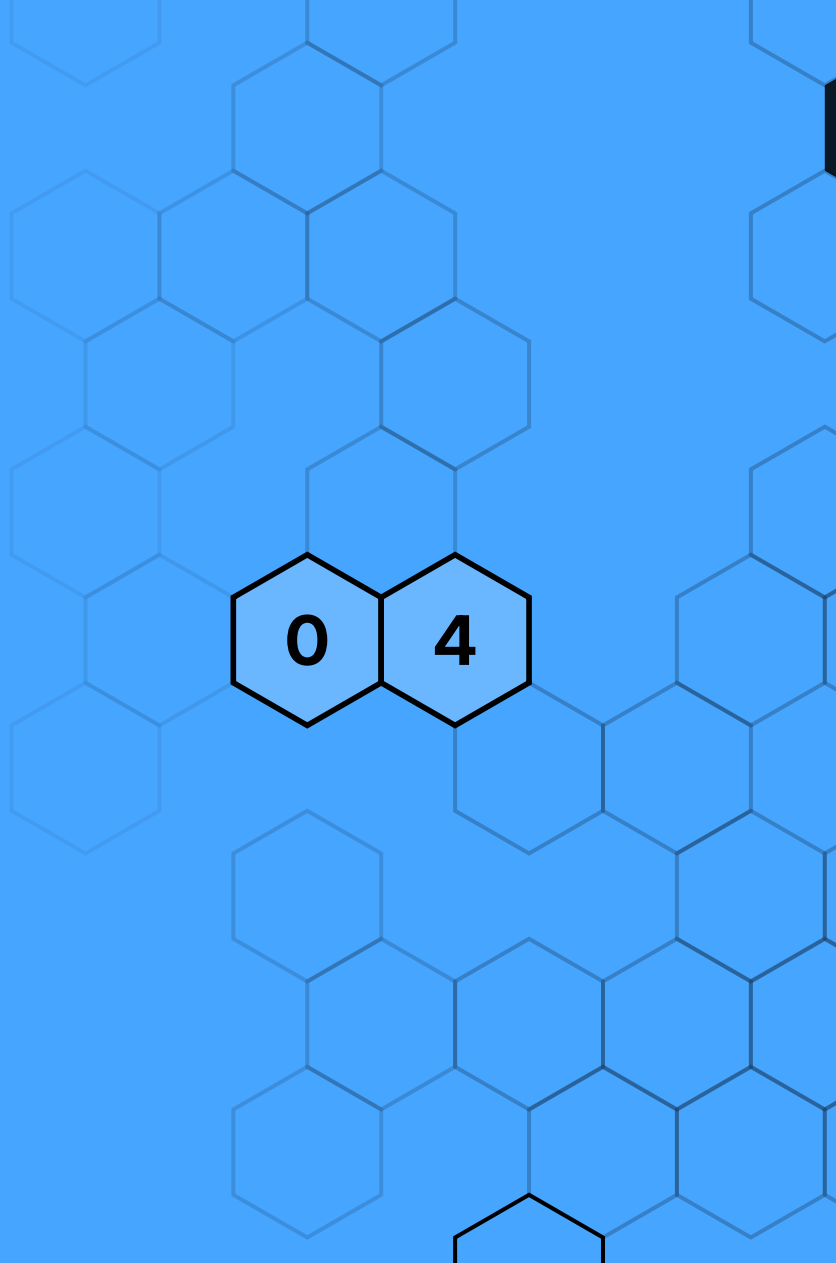



`filter_sketch.jpg`




**ImageGen**

# Hands-on example: using Blender



# The basic workflow

## 01 Design the prompt

Use a chatbot or AI agent to develop and refine your prompt.

## 02 Generate code, not images

Ask the AI to define shapes, structures, or physical setups in code. It can also work directly with your data.

## 03 Run it in Blender

Execute the generated code in Blender's scripting workspace.

## 04 Refine and render

Adjust the geometry, materials, and textures, then render the final result.

# The prompt

**PROMPT**

Create a Blender Python script that generates a scientific 3D visualization of microplastics in a water sample. Include a transparent blue water volume, hundreds of small irregular plastic fragments and thin fibers suspended at different depths, and a layer of sediment at the bottom. Use procedural geometry and materials, studio lighting, and an orthographic camera. Make the scene clean, visually clear, and suitable for an academic presentation. **Organize the parameters** so I can easily modify particles, sizes, shapes, and distribution. Generate the scene entirely through Python; do not use external images.

**AI**

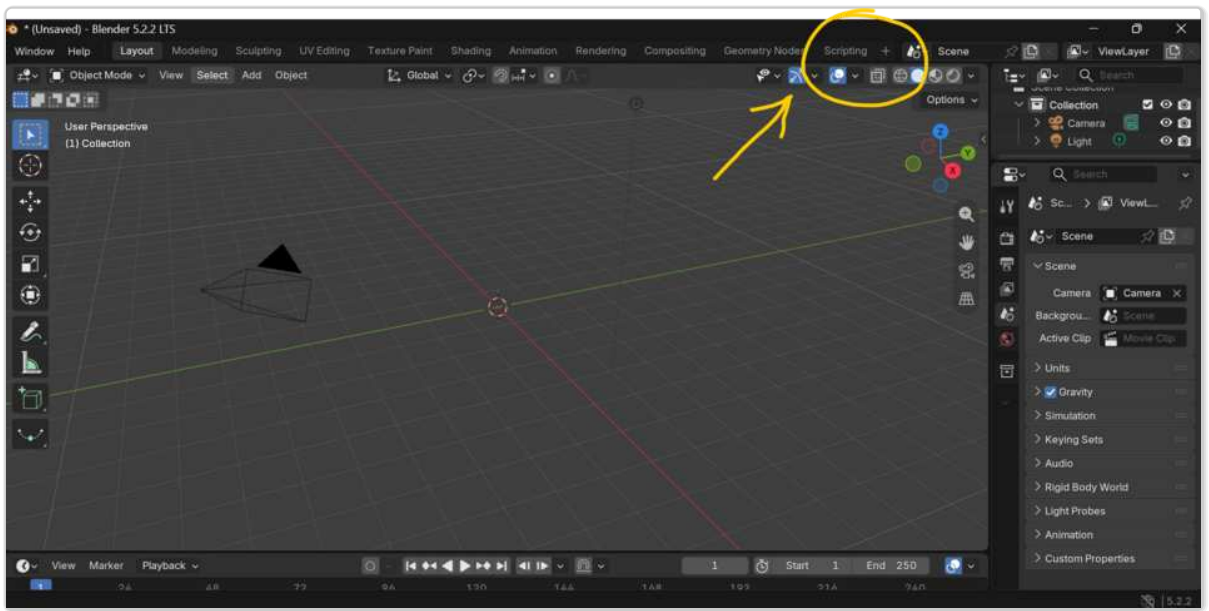
I've written the script as `microplastics_scene.py`. I haven't run it in Blender, so expect to adjust a few things on the first run. Run it from the Scripting tab, or headless with `blender --background --python microplastics_scene.py`.

### What it builds

- **Water:** a closed box with a transparent, Fresnel-reflective surface and a light-blue absorbing volume. Nothing refracts, so the particles stay undistorted.
- **Sediment:** a bumpy, noise-displaced layer at the bottom with a procedural sand material. About 700 loose sand grains sit on top of it.

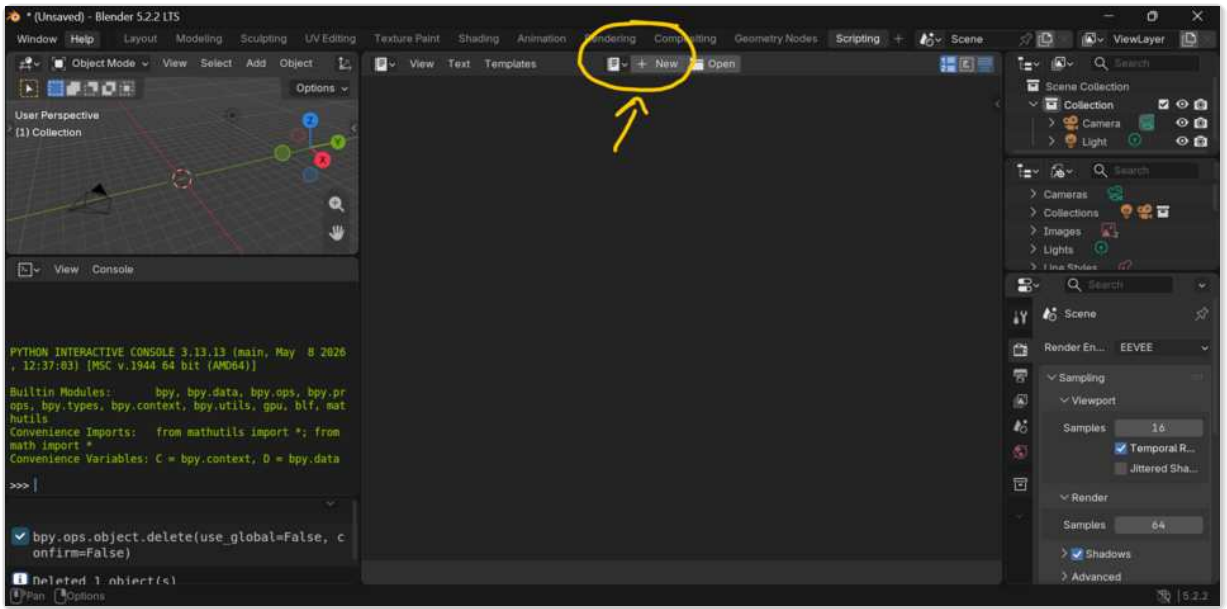
...

# Blender (how it looks)



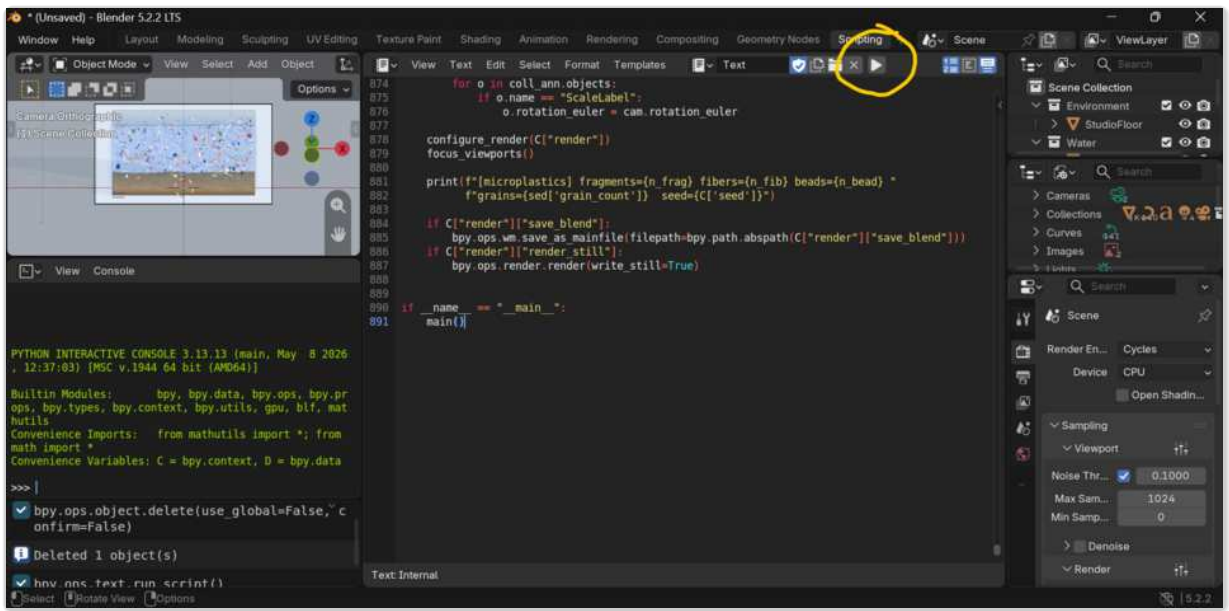

Go to **Scripting** workspace

# Blender (how to create a script)




Click on the **New** button

# Blender (how to run the script)




Click on the **Run** button

# Render



Check the result & iterate/save it

# Limits of AI-generated images

05

# Responsible use of AI in research

- It is just a new tool, we use it carefully
- **AI cannot be given authorship**
- Use of AI should be transparent by describing how AI is used (see example)
- Personal data protection (**GDPR**) is to be followed
- Protect your own unpublished work
- The output of the AI system can be sensitive
- When creating artistic output, contributor roles must be explained transparently and specifically.
- **Reproducibility** can be difficult to achieve
- **Accountability**: you are responsible for what you present

> Remember: the **AI Act is into force**. Research is exempt until it is put into practice.

# Copyrights (I)

## 1. Using other people's material as inputs to AI:

- Don't ask AI to "rework" someone else's copyrighted image/ text (derivative work)
- Style and mimicry: avoid asking for images that "closely resemble" a specific copyrighted work
- If you want to alter copyrighted work and then share it, you need the copyright holder's permission
- Allowed special cases: parody, caricature, pastiche (use carefully)
- Deepfakes: Must be clearly labelled as such (unethical deepfakes are still punishable!)

## 2. Using AI outputs:

- Ownership/protection: **If there's no real human creative input, the output isn't an "original work" in the copyright sense**. Your right to use it comes from the tool's license/ terms
- Risk of copying: AI might produce something too close to an existing copyrighted work (from training or your inputs). Avoid prompts like: "Make this look exactly like [living artist]/ [famous photo]"
- Provider responsibilities: Some providers add protections/ indemnities

# Copyrights (II)

## 3. Permanent copies made by the provider (training & storage)

- Check the tool's settings and terms
- Turn off "use my content for training" unless the material is yours, public-domain (author died >70 years ago), or explicitly CC0
- If the tool requires allowing training on uploads, don't upload third-party works.
- **If the provider stores your uploads for its own future use, you need the copyright owner's permission (unless it's your own/PD/CC0)**
- Also follow university standards: protect personal data, cite transparently, and follow good scientific practice

## 4. Using AI outputs in teaching and publications

- Don't publish AI outputs that rely on third-party, copyrighted inputs (e.g., publisher PDFs) as your teaching materials or figures
- You may publish AI outputs if the tool's terms allow it and you cite properly and the inputs were your own, public-domain, or CC0. Safe inputs for figures/materials: (1) your own images/diagrams; (2) Public domain (author died >70y ago); (3) CC0 content

# A!

## Kiitos!

**Pedro Santos Silva** · Aalto Scientific Computing

pedro.silva@aalto.fi

Enrico Glerean, Emrecan Gulay